Lecture 10

Two systems: tensor products and entanglement

Objectives. *A hydrogen atom in its ground state carries two spins, the electron's and the proton's, and they interact. This lecture gives the state space of a pair of systems: the tensor product of the two state spaces. A vector of the pair is described by a matrix of coefficients. The pair's state is a product of states of the parts exactly when this matrix has rank* 1. *Most states of a pair are not products, and such states are called* entangled. *The ground state of hydrogen's two spins is entangled, and an interaction can entangle two spins that start as a product. As a magnetic field grows, the ground state approaches a product state.*

By the end of this lecture you will be able to:

- Find the coefficient matrix of a state of a pair.
- Find the effect of a local operation on a state of a pair, using its coefficient matrix, and the evolution of two systems that do not interact, with constant Hamiltonians.
- Decide from the rank of the coefficient matrix whether a state of a pair is a product.
- Find the energies of hydrogen's two spins in a magnetic field along z, and the coefficient matrix of the ground state.

10.1 Two systems at once

Parts I and II treated one system at a time. Take now two systems, with state spaces $\mathbb{C}^n$ and $\mathbb{C}^m$: the spin of the electron and the spin of the proton in a hydrogen atom, say, each a qubit. What should the state space of the pair be? Three requirements come from experiment.

First, each system can be read alone. A reading of an observable A of the first system is still a reading, with the outcomes of A, when the second system is present.

Second, two systems prepared independently give independent readings. If the first is prepared in $[u]$ and the second in $[w]$, a reading a of the first and a reading b of the second should occur together with probability $\langle u, E_a u\rangle \langle w, G_b w\rangle$, the product of the two probabilities, where E_a and G_b are the projections of the two readings.

Third, the number of outcomes fixes the dimension. Read a complete observable of each system, one with n distinct readings and one with m. The pair then has nm distinguishable joint outcomes. As in § 4.1, an observable with nm distinct readings has nm orthonormal eigenvectors. So the pair's state space has dimension at least nm.

The *tensor product* of *Linear Algebra*, Def. 12.2, meets all three. The space $\mathbb{C}^n \otimes \mathbb{C}^m$ has the basis $e_i \otimes e_j$, where e_i runs over the standard basis of $\mathbb{C}^n$ and e_j over that of $\mathbb{C}^m$, so its dimension is nm. It has the inner product

$$\langle u \otimes w, u' \otimes w'\rangle = \langle u, u'\rangle \langle w, w'\rangle$$

(*Linear Algebra*, Rem. 12.7), in which the $e_i \otimes e_j$ are orthonormal. For matrices A on $\mathbb{C}^n$ and B on $\mathbb{C}^m$, the matrix $A \otimes B$ acts by $(A \otimes B)(u \otimes w) = Au \otimes Bw$ (*Linear Algebra*, Prop. 12.9). In particular the projection $E_a \otimes G_b$ gives

$$\langle u \otimes w, (E_a \otimes G_b)(u \otimes w)\rangle = \langle u, E_a u\rangle \, \langle w, G_b w\rangle \,,$$

which is the second requirement.

Postulate 10.1 (Two systems)**.**

1. *States.* Two systems with the state spaces $\mathbb{C}^n$ and $\mathbb{C}^m$ form a system with the state space $\mathbb{C}^n \otimes \mathbb{C}^m$.
2. *Local readings.* A reading of an observable A of the first system alone is a reading of the observable $A \otimes \mathbb{1}$ of the pair. A reading of an observable B of the second system alone is a reading of $\mathbb{1} \otimes B$.
3. *Independent preparations.* Two systems prepared independently in the states $[u]$ and $[w]$ form the pair in the state $[u \otimes w]$.
4. *No interaction.* Two systems that do not interact have the Hamiltonian $H_1 \otimes \mathbb{1} + \mathbb{1} \otimes H_2$, where H_1 and H_2 are their own Hamiltonians.

The three requirements motivate the postulate. They do not prove it, and it is a postulate, like those of Lecture 1 and Lecture 4. Everything else about the pair follows from those postulates applied to $\mathbb{C}^n \otimes \mathbb{C}^m$.

Every vector of the pair is a combination of the basis vectors,

$$\psi = \sum_{i=1}^{n} \sum_{j=1}^{m} C_{ij}\, e_i \otimes e_j, \tag{10.1}$$

and we call the $n \times m$ matrix C its *coefficient matrix*. The norm is $\|\psi\|^2 = \sum_{i,j} \left|C_{ij}\right|^2 = \mathrm{Tr}(C^\dagger C)$. A product has the coefficients $C_{ij} = u_i w_j$, that is, $C = uw^{\mathsf{T}}$, the column u times the row w^{T}. For two qubits we write $|jk\rangle = |j\rangle \otimes |k\rangle$, in the order $|00\rangle\,, |01\rangle\,, |10\rangle\,, |11\rangle$, in which the label of the first factor changes slowest. With the electron first, $|01\rangle$ is the electron up and the proton down.

10.2 Local readings and local evolution

The observables $A \otimes \mathbb{1}$ and $\mathbb{1} \otimes B$ commute, since both products equal $A \otimes B$ by *Linear Algebra*, Prop. 12.9. By Theorem 5.2 they are compatible, and their joint reading has the projections $E_a \otimes G_b$. So a reading of each system can be made at once, in either order, with the same joint statistics. If the pair is in the product state $[u \otimes w]$, the two readings are independent, by the computation before Postulate 10.1.

The coefficient matrix turns local operations into matrix products.

Proposition 10.2 (Local operations)**.** *Let ψ have the coefficient matrix C. For matrices A on $\mathbb{C}^n$ and B on $\mathbb{C}^m$, the vector $(A \otimes \mathbb{1})\psi$ has the coefficient matrix AC, the vector $(\mathbb{1} \otimes B)\psi$ has CB^{T}, and $(A \otimes B)\psi$ has ACB^{T}. Two systems that do not interact, with constant Hamiltonians H_1 and H_2, evolve by*

$$U_t = e^{-\mathrm{i}tH_1/\hbar} \otimes e^{-\mathrm{i}tH_2/\hbar}. \tag{10.2}$$

Proof. First, $(A \otimes \mathbb{1})\psi = \sum_{k,j} C_{kj}\, Ae_k \otimes e_j = \sum_{i,j} \big(\sum_k A_{ik} C_{kj}\big) e_i \otimes e_j$, and the bracket is $(AC)_{ij}$. In the same way, $(\mathbb{1} \otimes B)\psi = \sum_{i,j} \big(\sum_l C_{il} B_{jl}\big) e_i \otimes e_j$, and the bracket is $(CB^{\mathsf{T}})_{ij}$. The third statement follows, since $A \otimes B = (A \otimes \mathbb{1})(\mathbb{1} \otimes B)$.

Next, put $V_t = e^{-\mathrm{i}tH_1/\hbar} \otimes e^{-\mathrm{i}tH_2/\hbar}$. By the product rule,

$$\mathrm{i}\hbar \frac{\mathrm{d}V_t}{\mathrm{d}t} = H_1 e^{-\mathrm{i}tH_1/\hbar} \otimes e^{-\mathrm{i}tH_2/\hbar} + e^{-\mathrm{i}tH_1/\hbar} \otimes H_2 e^{-\mathrm{i}tH_2/\hbar} = (H_1 \otimes \mathbb{1} + \mathbb{1} \otimes H_2) V_t,$$

and $V_0 = \mathbb{1}$. The evolution U_t solves the same equation with the same initial value, so $U_t = V_t$ by the uniqueness of solutions (*Linear Algebra*, Prop. 6.21). ∎

The transpose in CB^{T} is not an adjoint. It appears because the second index of C belongs to the second system. For unitaries U and V, the state $(U \otimes V)\psi$ has the coefficient matrix UCV^{T}, and multiplying by the invertible matrices U and V^{T} does not change the rank. The next section shows what the rank measures.

10.3 Product and entangled states

Definition 10.3 (Product and entangled states). A state $[\psi]$ of the pair is a *product state* if $\psi = u \otimes w$ for some vectors $u \in \mathbb{C}^n$ and $w \in \mathbb{C}^m$. Otherwise it is *entangled*.

A product state is a pair of states, one for each system: the ray of u and the ray of w. An entangled state is a state of the pair that does not come from a state of each part.

Theorem 10.4 (The rank test). *A unit vector ψ with the coefficient matrix C is a product $u \otimes w$ if and only if* $\operatorname{rank} C = 1$. *For two qubits this holds if and only if* $\det C = 0$.

Proof. If $\psi = u \otimes w$, then $C = uw^{\mathsf{T}}$. Every column of C is a multiple of u, so $\operatorname{rank} C \le 1$, and $C \neq 0$ since $\psi \neq 0$. Hence $\operatorname{rank} C = 1$.

Conversely, let $\operatorname{rank} C = 1$. The columns of C span a line, spanned by some vector u, so the j-th column is $w_j u$ for a number w_j. Then $C_{ij} = u_i w_j$, and $\psi = \sum_{i,j} u_i w_j \, e_i \otimes e_j = u \otimes w$.

For two qubits, C is a nonzero 2×2 matrix, and its rank is 1 exactly when $\det C = 0$. ∎

Linear Algebra, Ex. 12.3 and Def. 17.11, previews this test. Nothing below depends on it. For two qubits the test is a single equation, $C_{00}C_{11} = C_{01}C_{10}$. A coefficient matrix chosen at random does not satisfy it. Most states of two qubits are entangled.

By Proposition 10.2, local unitaries $U \otimes V$ do not change the rank of the coefficient matrix. So two systems that do not interact, and evolve by (10.2), stay a product if they start as one, and stay entangled if they start entangled. Under evolution, only an interaction between the two systems can entangle them. A reading of an observable of the pair, not of either part alone, can entangle them too (Exercise Sheet 10). § 10.5 shows an interaction that entangles.

10.4 Correlation is not entanglement

The state

$$\psi^- = \frac{1}{\sqrt{2}}\big(|01\rangle - |10\rangle\big)$$

has the coefficient matrix $C = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, with $\det C = \frac{1}{2} \neq 0$, so, by Theorem 10.4, it is entangled. Read both spins along z. The joint projections are $E_s \otimes E_t$, $s, t \in \{+, -\}$, with $E_\pm = \frac{1}{2}(\mathbb{1} \pm \sigma_z)$, and the two readings are always opposite. The outcomes are $\uparrow\downarrow$ and $\downarrow\uparrow$, each with probability $\frac{1}{2}$.

Opposite readings along z do not need entanglement. Take the ensemble of pairs $\{(\frac{1}{2}, |01\rangle), (\frac{1}{2}, |10\rangle)\}$, prepared by tossing a coin and preparing the product $|01\rangle$ or $|10\rangle$. Its density matrix

$$\rho = \tfrac{1}{2}\big(|01\rangle\langle 01| + |10\rangle\langle 10|\big)$$

gives the same opposite readings along z, with the same probabilities. The two preparations differ along x. Since $(\sigma_x \otimes \sigma_x)\psi^- = -\psi^-$, the mean of the product of the two readings along x is -1 in ψ^-. So the readings along x are again opposite, with certainty. In the ensemble the mean is $\operatorname{Tr}\big(\rho\,(\sigma_x \otimes \sigma_x)\big) = 0$, and the two readings along x agree with probability $\frac{1}{2}$.

No ensemble of product states gives opposite readings with certainty along both axes. For a product $u \otimes w$, with Bloch vectors $\mathbf{r}$ of u and $\mathbf{s}$ of w, the means of the products along z and along x add up to

$$\langle u \otimes w, (\sigma_z \otimes \sigma_z + \sigma_x \otimes \sigma_x)\, u \otimes w\rangle = r_z s_z + r_x s_x \geq -1,$$

by the Cauchy–Schwarz inequality for the vectors (r_z, r_x) and (s_z, s_x), whose lengths are at most 1. The mean in an ensemble is $\operatorname{Tr}(\rho A)$, by (3.6), and for an ensemble of products this is the weighted average of the members' means. So it is also at least -1. In ψ^- the sum is -2.

So correlation along one axis is not entanglement. Neither is entanglement a property of a basis. The vectors $|01\rangle$, $t_0 = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ and ψ^- are all eigenvectors of $\sigma_z \otimes \sigma_z$ with the eigenvalue -1. The first is a product, and the other two are entangled.

10.5 The two spins of hydrogen

In a hydrogen atom the electron's spin and the proton's spin interact through their magnetic moments. Take the electron in its lowest orbital, keep it there, and apply no magnetic field. Then only the two spins can change. They form two qubits, and, once the constant energy of the orbital is dropped, their effective Hamiltonian is

$$H = \frac{\hbar\nu}{4}\,\boldsymbol{\sigma}\cdot\boldsymbol{\sigma}, \qquad \boldsymbol{\sigma}\cdot\boldsymbol{\sigma} = \sigma_x \otimes \sigma_x + \sigma_y \otimes \sigma_y + \sigma_z \otimes \sigma_z, \tag{10.3}$$

where $\nu > 0$ is an angular frequency, measured as $\nu/2\pi = 1\,420.405\,751\,768$ MHz. To find its levels we use the *swap* W on $\mathbb{C}^2 \otimes \mathbb{C}^2$, the matrix with $W(u \otimes w) = w \otimes u$.

Lemma 10.5 (The swap). *On two qubits, $\boldsymbol{\sigma}\cdot\boldsymbol{\sigma} = 2W - \mathbb{1}$.*

Proof. Both sides are matrices, so it is enough to compare them on the four basis vectors. By the Pauli matrices of Lecture 1, $(\sigma_z \otimes \sigma_z)\,|00\rangle = |00\rangle$, $(\sigma_x \otimes \sigma_x)\,|00\rangle = |11\rangle$ and $(\sigma_y \otimes \sigma_y)\,|00\rangle = \mathrm{i}\cdot\mathrm{i}\,|11\rangle = -|11\rangle$, so $\boldsymbol{\sigma}\cdot\boldsymbol{\sigma}\,|00\rangle = |00\rangle = (2W - \mathbb{1})\,|00\rangle$. Next, $(\sigma_z \otimes \sigma_z)\,|01\rangle = -|01\rangle$, $(\sigma_x \otimes \sigma_x)\,|01\rangle = |10\rangle$ and $(\sigma_y \otimes \sigma_y)\,|01\rangle = \mathrm{i}\cdot(-\mathrm{i})\,|10\rangle = |10\rangle$, so $\boldsymbol{\sigma}\cdot\boldsymbol{\sigma}\,|01\rangle = 2\,|10\rangle - |01\rangle = (2W - \mathbb{1})\,|01\rangle$. The vectors $|11\rangle$ and $|10\rangle$ follow in the same way, with 0 and 1 exchanged. ∎

The swap is self-adjoint and $W^2 = \mathbb{1}$, so its eigenvalues are ± 1. The eigenspace of $+1$ is the space of *symmetric* vectors, with the orthonormal basis $|00\rangle$, t_0, $|11\rangle$. The eigenspace of -1 is the line of ψ^-, the *antisymmetric* vectors. These are the spaces $S^2\mathbb{C}^2$ and $\Lambda^2\mathbb{C}^2$ of *Linear Algebra*, Def. 12.20. For two spins the three symmetric vectors are called the *triplet*, and ψ^- the *singlet*.

Example 10.6 (Hydrogen's two spins). Figure 10.1 shows the coefficient matrices that occur.

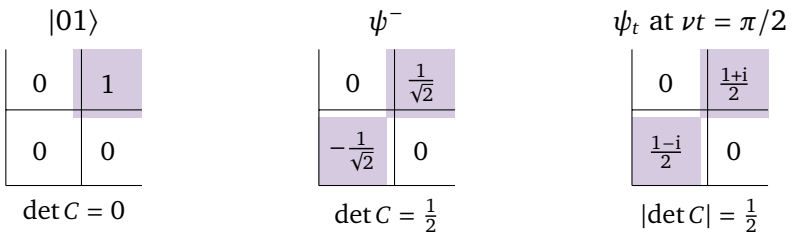


Figure 10.1: Coefficient matrices of three states of two qubits; rows are the first factor, columns the second. The product $|01\rangle$ has a matrix of rank 1. The singlet ψ^- and the state ψ_t of Example 10.6 at $\nu t = \pi/2$ (without its phase $e^{-i\nu t/4}$) have rank 2. Shaded entries are not zero.

1. *Levels.* By Lemma 10.5, $H = \frac{\hbar\nu}{4}(2W - \mathbb{1})$. It has the energy $\hbar\nu/4$ on the three symmetric vectors and $-3\hbar\nu/4$ on ψ^-. The lowest state of hydrogen's two spins is ψ^-, which is entangled. The two levels are $\hbar\nu$ apart. The light emitted between them is the 21 cm line of hydrogen.
2. *A product evolves.* Prepare the electron up and the proton down, $|01\rangle = \frac{1}{\sqrt2}(t_0 + \psi^-)$. By Theorem 4.4 each of the two components gains its own phase:

$$\psi_t = \frac{1}{\sqrt2}\left(e^{-i\nu t/4}t_0 + e^{3i\nu t/4}\psi^-\right) = e^{-i\nu t/4}\left(\frac{1+e^{i\nu t}}{2}\,|01\rangle + \frac{1-e^{i\nu t}}{2}\,|10\rangle\right).$$

3. *The spins trade orientations.* Reading both spins along z gives $\uparrow\downarrow$ with probability $\left|(1+e^{i\nu t})/2\right|^2 = \cos^2(\nu t/2)$ and $\downarrow\uparrow$ with probability $\sin^2(\nu t/2)$. The electron's and the proton's orientations are exchanged and exchanged back, at the Bohr frequency ν of Lecture 4.
4. *The interaction entangles.* The vector $e^{i\nu t/4}\psi_t$ describes the same state, and its coefficient matrix has the determinant

$$-\frac{(1+e^{i\nu t})(1-e^{i\nu t})}{4} = -\frac{1-e^{2i\nu t}}{4}, \qquad |\det C_t| = \frac{|\sin \nu t|}{2}.$$

   The state is a product only when νt is a multiple of π. At all other times the two spins are entangled.

10.6 Hydrogen in a magnetic field

A magnetic field of magnitude B along z acts on each spin by its own Larmor term (Lecture 1). The electron's gyromagnetic ratio is negative, of size γ_e (Lecture 6), and the proton's, γ_p, is positive, so with the angular frequencies $\nu_e = \gamma_e B$ and $\nu_p = \gamma_p B$,

$$H = \frac{\hbar\nu}{4}\,\boldsymbol{\sigma}\cdot\boldsymbol{\sigma} + \frac{\hbar\nu_e}{2}\,\sigma_z\otimes\mathbb{1} - \frac{\hbar\nu_p}{2}\,\mathbb{1}\otimes\sigma_z. \tag{10.4}$$

For one tesla, $\nu_e/2\pi = 28.0$ GHz and $\nu_p/2\pi = 42.6$ MHz.

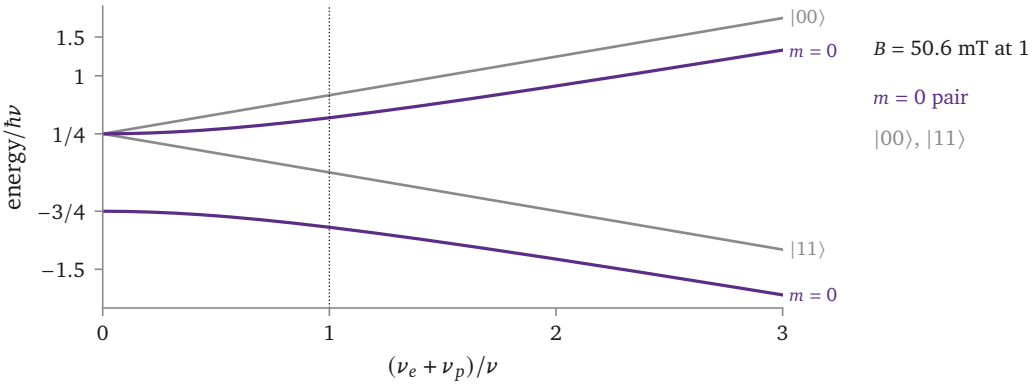


Figure 10.2: The four energies of Proposition 10.7 in units of $\hbar\nu$, against $(\nu_e+\nu_p)/\nu$; dotted: $B = 50.6$ mT.

Proposition 10.7 (Hydrogen's two spins in a field). *Let $\Sigma_z = \sigma_z \otimes \mathbb{1} + \mathbb{1} \otimes \sigma_z$, and let H be (10.4).*

1. *Σ_z commutes with H. Its eigenvalues are 2 on $|00\rangle$, -2 on $|11\rangle$ and 0 on the plane spanned by $|01\rangle$ and $|10\rangle$, and H maps each of these eigenspaces into itself.*
2. *$|00\rangle$ and $|11\rangle$ are eigenvectors of H, with the energies $\hbar\nu/4 \pm \hbar(\nu_e - \nu_p)/2$.*
3. *On the plane, in the basis $|01\rangle$, $|10\rangle$, H has the matrix*

$$\begin{pmatrix} -\hbar\nu/4 + \hbar(\nu_e+\nu_p)/2 & \hbar\nu/2 \\ \hbar\nu/2 & -\hbar\nu/4 - \hbar(\nu_e+\nu_p)/2 \end{pmatrix}$$

and the energies $-\hbar\nu/4 \pm \frac{\hbar}{2}\sqrt{\nu^2 + (\nu_e+\nu_p)^2}$. The lower of the two is the lowest energy of H, for every field.

Proof. By Lemma 10.5 the coupling is $\frac{\hbar\nu}{4}(2W - \mathbb{1})$. The swap commutes with Σ_z, since it exchanges its two terms, and the field terms are diagonal, so (i) holds. By Lemma 4.3, H keeps the eigenspaces of Σ_z. The swap fixes $|00\rangle$ and $|11\rangle$, which gives (ii). It exchanges $|01\rangle$ and $|10\rangle$, which gives the off-diagonal entries $\hbar\nu/2$ in (iii). The diagonal entries are $-\hbar\nu/4$ plus the field terms. Proposition 6.7 gives the two energies. The lower one is at most $-\hbar\nu/4 - \hbar(\nu_e+\nu_p)/2$, which lies below the energy $\hbar\nu/4 - \hbar(\nu_e-\nu_p)/2$ of $|11\rangle$ by $\hbar\nu/2 + \hbar\nu_p$. The energy of $|00\rangle$ is no lower, since $\nu_e \geq \nu_p$. ∎

We write $m = 1, 0, -1$ for the eigenvalues of $\Sigma_z/2$, the pair's total spin along z in units of $\hbar$ (Lecture 13). At $B = 0$ the energies are those of Example 10.6, and Figure 10.2 follows them as the field grows.

Example 10.8 (The ground state in a field). The ground state is the eigenvector of the matrix of H on the plane (Proposition 10.7) for its lower energy. Write $a = \hbar(\nu_e+\nu_p)/2$, $\upsilon = \hbar\nu/2$ and $r = \sqrt{a^2+\upsilon^2}$, so that the matrix is $-\hbar\nu/4$ plus $\left(\begin{smallmatrix} a & \upsilon \\ \upsilon & -a \end{smallmatrix}\right)$. Take the unit eigenvector $c_1|01\rangle + c_2|10\rangle$ for $-r$ with $c_1 > 0$. By the first row, $c_2 = -(a+r)c_1/\upsilon$, which is real. So $c_1^2 = \upsilon^2/\big(\upsilon^2 + (a+r)^2\big) = \upsilon^2/\big(2r(a+r)\big)$ and $c_1c_2 = -\upsilon/(2r)$. Its coefficient matrix has $\det C = -c_1c_2$, so

$$|\det C| = \frac{\nu}{2\sqrt{\nu^2 + (\nu_e+\nu_p)^2}}.$$

This is $\frac{1}{2}$ at $B = 0$, and $1/(2\sqrt{2})$ at $\nu_e + \nu_p = \nu$, that is, at $B = 50.6$ mT. It tends to 0 as the field grows, though it is positive for every field. So by Theorem 10.4 the ground state is entangled at every field. In a strong field it is nearly the product $|10\rangle$: the electron down, the proton up, each magnetic moment along the field.

Lecture 11 defines a measure of how entangled a pure pair is, and for two qubits this measure grows with $|\det C|$.

Each field term acts on one spin, yet the ground state approaches a product as B grows. This does not contradict Proposition 10.2, which is about evolution under one Hamiltonian. Each B gives another Hamiltonian, with its own ground state. Moreover, in a field t_0 and ψ^- are no longer eigenvectors, so the triplet and the singlet are the levels at $B = 0$ only.

The physics

The interaction (10.3) splits the lowest state of hydrogen into a triplet and a singlet, $\hbar\nu$ apart, with $\nu/2\pi \approx 1420$ MHz, which is a wavelength of 21 cm. Van de Hulst predicted in 1944 that interstellar hydrogen should emit this line, and Ewen and Purcell observed it in 1951. An atom in the triplet, left alone, falls to the singlet after about eleven million years on average. Clouds of hydrogen are so large that the line is bright, and radio astronomers map the hydrogen of the galaxy with it. In such a cloud the atoms are a mixture of the four states, not the singlet. Collisions between atoms keep the triplet populated. In the laboratory the hydrogen maser holds atoms in the triplet and makes them emit. A maser runs on the line between the two $m = 0$ levels of § 10.6, whose spacing is

$$\hbar\sqrt{\nu^2 + (\nu_e + \nu_p)^2} = \hbar\nu + \hbar(\nu_e + \nu_p)^2/(2\nu) + \cdots .$$

Near zero field, a stray field therefore moves it only at second order, by 0.28 Hz at 1 μT. Breit and Rabi found the four levels in 1931. Measurements with hydrogen masers, corrected for the masers' own small shifts, give the frequency of the transition as 1 420 405 751.768 Hz, one of the most precisely measured numbers in physics.

Two systems

State space: $\mathbb{C}^n \otimes \mathbb{C}^m$; local readings are $A \otimes \mathbb{1}$ and $\mathbb{1} \otimes B$ (Postulate 10.1). *Coefficients:* $\psi = \sum C_{ij} e_i \otimes e_j$; $(U \otimes V)\psi$ has UCV^{T} (Proposition 10.2). *Entanglement:* ψ is a product iff $\operatorname{rank} C = 1$; two qubits: $\det C = 0$ (Theorem 10.4). *Hydrogen:* $H = \frac{\hbar\nu}{4}\boldsymbol{\sigma}\cdot\boldsymbol{\sigma} = \frac{\hbar\nu}{4}(2W - \mathbb{1})$; the ground state ψ^- is entangled (Example 10.6). In a field, $|00\rangle$ and $|11\rangle$ stay eigenvectors and the $m = 0$ pair is a 2×2 problem (Proposition 10.7).

Summary.

- *Two systems*: the state space is the tensor product, and a reading of one system is a reading of $A \otimes \mathbb{1}$ (Postulate 10.1).
- *The rank test*: a state is a product exactly when its coefficient matrix has rank 1 (Theorem 10.4).
- *Hydrogen's two spins*: their ground state ψ^- is entangled, and their interaction entangles a product state (Example 10.6).
- *In a field*: the ground state tends to a product as the field grows (Example 10.8).

Lecture 11 asks what a reading of one of the two spins sees when the pair is entangled.

References

Nielsen and Chuang, *Quantum Computation and Quantum Information*, §2.2.8, pp. 93–96. Cohen-Tannoudji, Diu and Laloë, *Quantum Mechanics*, vol. 1, Ch. II §F, pp. 147–157, and vol. 2, Ch. XII, §§D–E, pp. 1246–1264. *Linear Algebra for Physics*, Def. 12.2, Prop. 12.9, Rem. 12.7, Def. 12.20. Ewen and Purcell, *Nature* 168, 356 (1951). Breit and Rabi, *Phys. Rev.* 38, 2082 (1931).